Name: ______________________

Class: ______________________

Date: ______________________

Mr. Rawson • GCSE Physics

REVIEW — Series Resistors

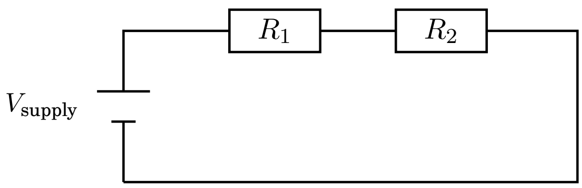
Keep this sheet. Everything you need for series circuits is on this side.

Key concept

THE THREE RULES — series (one loop, no choice of route)

1. **Current is the same everywhere.** $I_1 = I_2 = I_{\text{supply}}$
   Charge is not used up. An ammeter reads the same wherever you put it in the loop.
2. **Potential difference is shared out.** $V_{\text{supply}} = V_1 + V_2$
   The bigger resistor takes the bigger share.
3. **Resistances add.** $\boxed{R_{\text{total}} = R_1 + R_2}$
   Adding a resistor in series always *increases* total resistance — the charge must pass through every one of them, so there is more opposition on the single route available.

And $V = IR$ still applies to *every single component*, as well as to the whole circuit.



same current all the way round

Worked example

WORKED EXAMPLE 1 — the standard one

A $4\,\Omega$ and a $6\,\Omega$ resistor are connected in series to a $12\,\text{V}$ supply. Find the current, and the potential difference across each resistor.

1. Total resistance $R_{\text{total}} = R_1 + R_2 = 4 + 6 = 10\,\Omega$

2. Current $I = \dfrac{V}{R} = \dfrac{12}{10} = 1.2\,\text{A}$ (this is the current *everywhere*)

3. Each p.d. $V_1 = IR_1 = 1.2 \times 4 = 4.8\,\text{V}$
$V_2 = IR_2 = 1.2 \times 6 = 7.2\,\text{V}$

4. Check $4.8 + 7.2 = 12\,\text{V}$ ✓ matches the supply — rule 2

Worked example

WORKED EXAMPLE 2 — working backwards

A $9\,\text{V}$ supply drives a current of $0.30\,\text{A}$ through two resistors in series. One of them is $12\,\Omega$. Find the other.

1. Total resistance $R_{\text{total}} = \dfrac{V}{I} = \dfrac{9}{0.30} = 30\,\Omega$

2. Subtract $R_2 = R_{\text{total}} - R_1 = 30 - 12 = 18\,\Omega$

AQA Trilogy 8464 · spec references aP2-18, aP2-19, aP2-22, aP2-24, aP2-25

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REVIEW — Series Resistors: practice

1) **State** what happens to the size of the current as you move an ammeter to different points around a series circuit. **[1]**

2) **Calculate** the total resistance of a $3\,\Omega$ and a $7\,\Omega$ resistor in series. **[1]**

3) **Calculate** the current when those two resistors are connected to a $20\,\mathrm{V}$ supply. **[2]**

4) **Calculate** the potential difference across each of the two resistors. **[2]**

5) **Show** that your two answers above are consistent with the supply potential difference. **[1]**

6) **Calculate** the unknown resistance: a $12\,\mathrm{V}$ supply drives $0.40\,\mathrm{A}$ through two resistors in series, one of which is $10\,\Omega$. **[3]**

7) **Explain** qualitatively why adding another resistor in series *increases* the total resistance of a circuit. **[2]**

8) **Explain** what happens to the other lamp when one lamp in a series circuit is removed, and why. **[2]**